Quantum Edge Detection Under Changing Image Conditions

Final Report — QPIE-QHED vs NEQR-Gradient on Grayscale Lunar Crater Imagery

Generated from results/tables on 2026-10-04 17:26. All quantum numbers are mean ± std over n=10 trials (shot-seed variation); classical methods are deterministic. Thresholds are validation-selected (results/tables/thresholds.csv). Ground truth is independent (analytic / annotation-derived masks) — Sobel and Canny are baselines, never the reference.

1. Abstract

We implemented two quantum image representations for edge detection — QPIE amplitude encoding with the Quantum Hadamard Edge Detection (QHED) algorithm, and NEQR basis-state encoding with an exact quantum subtractor-based gradient operator — and compared them with classical Sobel, Canny, and 8-direction Robinson baselines on synthetic geometries and lunar crater imagery under image degradations and simulated hardware noise. Both quantum circuits were verified against analytic target statevectors to machine precision before benchmarking. Headline findings: (i) both representations recover comparable edge quality on synthetic images, but NEQR degrades far more gracefully on real lunar patches; (ii) NEQR's resource cost grows ~20x faster in gates than QHED's despite both adding qubits logarithmically; (iii) sampling variance follows the 1/shots law with F1 saturation by ~1,024 shots; (iv) QHED's amplitude normalisation makes it contrast-sensitive in a way NEQR is not; (v) under equal depolarizing rates the deeper NEQR circuit loses fidelity faster (H5 confirmed).

2. Experiment 1 — Representation ablation

|:----------|:---------|--------------:|-------------:|---------:|------:|----------:|

Resources (8x8, ideal):

|:----------|:---------------------------------|:---------|-----------------:|----------:|---------------:|----------------:|-------------:|

Key result: on synthetic two-level images both quantum representations reach the same F1 (the binary maps coincide at the selected threshold); on real lunar patches they diverge sharply — NEQR retains high F1 while QHED's $P \propto |\Delta a|^2$ mapping compresses the weak amplitude differences of low-contrast terrain.

3. Experiment 2 — Resolution scaling

|:-----------|-------:|:---------|--------------:|---------------:|------------:|--------------------:|--------------------:|----------------:|-------------:|

Logical qubits grow logarithmically (QHED 5→7→9; NEQR 22→24→26) but gate counts grow linearly-to-quadratically in pixels (NEQR 289→1,301→6,009), confirming H6: total resource cost is dominated by gate/depth scaling, not qubit count.

4. Experiment 3 — Shot-count sensitivity

|:---------|--------:|--------------:|-------------:|--------------:|-------------:|----------------:|

Per-pixel map variance scales as 1/shots (binomial law) and boundary F1 saturates by ~1,024–2,048 shots; 512-shot runs show elevated variance (H2 confirmed with diminishing returns above saturation).

5. Experiment 4 — Image degradation robustness

Robustness AUC (area under F1-vs-degradation curve):

|:---------------|:--------|:---------|---------:|--------:|--------:|-----------:|--------:|--------:|

Sobel is the most degradation-tolerant; the quantum methods lose most under contrast scaling (QHED) and blur. QHED's amplitude normalisation makes global contrast changes affect edge probabilities non-linearly — a representation-intrinsic sensitivity absent in NEQR.

6. Experiment 5 — Quantum hardware noise

|:---------|-------:|-----------------:|:-------------|--------:|--------------:|-------------:|----------------:|---------------:|

Readout noise corrupts all methods roughly equally; depolarizing noise hurts the deeper NEQR circuit more at equal rates (H5), and coherent over-rotation injects systematic bias that neither stochastic model captures.

7. Experiment 6 — Geometry ablation

|:----------------|-------:|-------:|

Failure classification (Robinson diagnostic):

|:----------------|:---------|-------------:|---------------:|-------------------------:|:--------------------|

Robinson dominant-direction diagnostics separate quantum-specific failures (Case A: classical detectors respond strongly, quantum F1 drops) from image-level failures (Case B: everything struggles).

8. Experiment 7 — Preprocessing

|:-------------|--------:|-------:|-------:|-----------:|--------:|

9. Experiment 8 — Lunar crater benchmark

|-------:|:---------|-----------:|---------:|---------:|------------:|---------:|------:|----------:|---------:|

10. Experiment 9 — Resource scaling

|-------:|:---------|-----------------:|----------:|---------------:|------------:|---------:|---------:|------------:|-------------------:|----------------:|---------------------:|-------------------:|-----------------:|-------------------:|--------------:|----------------:|-------------:|-------------------:|

This project evaluates edge detection quality and robustness, not quantum computational advantage. State preparation, measurement, and reconstruction dominate total quantum pipeline time at these small scales.

11. Experiment 10 — Failure taxonomy

|:-------------------------|--------:|------------:|-------------------:|------------:|-----------:|--------:|

Severity classification:

|:-------------------------|:----------------|:-----------|

Visual galleries: results/failure_gallery/.

12. Hypothesis scorecard

|----|-----------|---------|

13. Conclusions and future work

1. **Representation matters.** The choice of quantum image representation changes both the physics of the edge operator (Hadamard interference vs quantum arithmetic) and the robustness profile on real imagery. NEQR's exact integer differences transfer better to low-contrast planetary terrain than QHED's quadratic amplitude mapping.
2. **Resources, not qubits, are the bottleneck.** Both representations add qubits logarithmically, but NEQR's multi-controlled preparation and ripple-carry arithmetic make its gate count and depth grow orders of magnitude faster; at 16x16 the NEQR circuit is ~6,000 gates deep-complex vs QHED's ~30.
3. **Measurement budget saturates early.** 1,024-2,048 shots capture nearly all attainable edge quality in the ideal setting; more shots only shrink binomial tails.
4. **Noise asymmetry.** Readout error is representation-agnostic; depolarizing noise amplifies with circuit depth, penalising NEQR; coherent over-rotation biases both but is invisible to averaged fidelity metrics.
5. **Classical baselines remain far cheaper** and more robust at these resolutions; the

quantum value proposition at NISQ scales remains exploratory, concentrated in representation-level operators (exact quantum arithmetic on encoded images) rather than end-to-end speed.

Future work: bigger patches via blockwise QHED, FAQUAM/other encodings, real-hardware proof-of-concept on IBM Quantum (4x4 only), and learned quantum circuit compilation for the subtractor network.

Appendix A — Reproducibility

```
python tests/test_gates.py && python tests/test_qhed.py && python tests/test_neqr.py
python experiments/run_all.py          # all 10 experiments
python report/build_report.py          # this report
streamlit run dashboard/app.py         # interactive dashboard
```

Seeds: global seed 42, trial seeds 42..51, shot seeds = trial seeds. Simulator: Qiskit AerSimulator (ideal: statevector; noisy: statevector trajectories with hardware-basis transpilation {rz, sx, x, cx} so gate-level noise applies).

Appendix B — Figures

Tables

dataset	method	f1_1px_mean	f1_1px_std	pr_auc	iou	chamfer
lunar	canny	0.223	0.245	0.865	0.112	4.240
lunar	neqr	0.881	0.080	0.831	0.795	0.380
lunar	qhed	0.418	0.111	0.808	0.221	0.777
lunar	robinson	0.880	0.080	0.882	0.794	0.383
lunar	sobel	0.870	0.078	0.875	0.765	0.385
synthetic	canny	0.858	0.081	0.678	0.432	0.540
synthetic	neqr	0.900	0.095	0.599	0.466	0.416
synthetic	qhed	0.900	0.095	0.825	0.466	0.416
synthetic	robinson	0.938	0.091	0.654	0.824	0.559
synthetic	sobel	0.933	0.094	0.718	0.810	0.579

dataset	image	method	logical_qubits	ancilla	total_qubits	kernel_time_s	e2e_time_s
synthetic	horizontal_line	qhed	6	1	7	0.029	0.209
synthetic	horizontal_line	neqr	22	2	24	0.000	0.016
lunar	patch_8px_lunarfb_000_169_82.png	qhed	6	1	7	0.030	0.231
lunar	patch_8px_lunarfb_000_169_82.png	neqr	22	2	24	0.000	0.027

geometry	size	method	f1_1px_mean	total_qubits	raw_gates	transpiled_gates	transpiled_depth	kernel_time_s	e2e_time_s
circle	4	qhed	0.933	5	18	29	17	0.026	0.210
circle	4	neqr	0.933	22	290	245	184	0.000	0.005
circle	8	qhed	0.776	7	23	116	98	0.030	0.227
circle	8	neqr	0.776	24	1302	805	648	0.000	0.014
circle	16	qhed	0.864	9	28	356	333	0.039	0.284
circle	16	neqr	0.864	26	6010	2949	2408	0.001	0.051
ellipse	4	qhed	0.933	5	18	29	17	0.027	0.220
ellipse	4	neqr	0.933	22	290	245	184	0.000	0.005
ellipse	8	qhed	0.895	7	23	115	97	0.029	0.233
ellipse	8	neqr	0.895	24	1286	789	632	0.000	0.014
ellipse	16	qhed	0.909	9	28	481	458	0.041	0.305
ellipse	16	neqr	0.909	26	5962	2901	2360	0.001	0.049

method	shots	f1_1px_mean	f1_1px_std	pr_auc_mean	pr_auc_std	map_variance
qhed	512	0.7590	0.0214	0.9181	0.0070	0.0187
qhed	1024	0.7755	0.0000	0.9174	0.0081	0.0067
qhed	2048	0.7755	0.0000	0.9213	0.0000	0.0032
qhed	4096	0.7755	0.0000	0.9213	0.0000	0.0017
qhed	8192	0.7755	0.0000	0.9213	0.0000	0.0008
qhed	16384	0.7755	0.0000	0.9213	0.0000	0.0004
qhed	32768	0.7755	0.0000	0.9213	0.0000	0.0002
neqr	512	0.7755	0.0000	0.8444	0.0000	0.0000
neqr	1024	0.7755	0.0000	0.8444	0.0000	0.0000
neqr	2048	0.7755	0.0000	0.8444	0.0000	0.0000
neqr	4096	0.7755	0.0000	0.8444	0.0000	0.0000
neqr	8192	0.7755	0.0000	0.8444	0.0000	0.0000
neqr	16384	0.7755	0.0000	0.8444	0.0000	0.0000
neqr	32768	0.7755	0.0000	0.8444	0.0000	0.0000

family	level	method	f1_1px	sobel	canny	robinson	qhed	neqr
gaussian_noise	AUC	nan	nan	0.977	0.831	0.892	0.776	0.774
gaussian_blur	AUC	nan	nan	0.948	0.833	0.934	0.879	0.885
contrast	AUC	nan	nan	1.000	0.833	0.875	0.776	0.776
illumination	AUC	sobel	0.977	nan	nan	nan	nan	nan
illumination	AUC	canny	0.833	nan	nan	nan	nan	nan
illumination	AUC	robinson	0.904	nan	nan	nan	nan	nan
illumination	AUC	qhed	0.789	nan	nan	nan	nan	nan
illumination	AUC	neqr	0.789	nan	nan	nan	nan	nan

method	size	intensity_bits	noise	shots	f1_1px_mean	f1_1px_std	fidelity_mean	fidelity_std
qhed	8	8	ideal	2048	0.776	0.000	0.980	0.008
qhed	8	8	depol_0.001	2048	0.958	0.014	0.926	0.005
qhed	8	8	depol_0.005	2048	0.904	0.004	0.749	0.009
qhed	8	8	depol_0.01	2048	0.897	0.002	0.605	0.007
qhed	8	8	depol_0.05	2048	0.897	0.000	0.373	0.012

method	size	intensity_bits	noise	shots	f1_1px_mean	f1_1px_std	fidelity_mean	fidelity_std
qhed	8	8	readout_0.01	2048	0.926	0.025	0.965	0.008
qhed	8	8	readout_0.05	2048	0.945	0.017	0.884	0.009
qhed	8	8	readout_0.10	2048	0.913	0.014	0.784	0.012
qhed	8	8	overrot	2048	0.866	0.017	0.968	0.004
qhed	8	8	combined	2048	0.899	0.005	0.681	0.012
qhed	2	8	ideal	2048	1.000	0.000	0.999	0.001
qhed	2	8	depol_0.001	2048	1.000	0.000	0.999	0.001
qhed	2	8	depol_0.005	2048	1.000	0.000	0.999	0.001
qhed	2	8	depol_0.01	2048	1.000	0.000	0.999	0.001
qhed	2	8	depol_0.05	2048	1.000	0.000	0.996	0.001
qhed	2	8	readout_0.01	2048	1.000	0.000	0.999	0.001
qhed	2	8	readout_0.05	2048	1.000	0.000	0.999	0.001
qhed	2	8	readout_0.10	2048	1.000	0.000	0.998	0.001
qhed	2	8	overrot	2048	1.000	0.000	0.999	0.001
qhed	2	8	combined	2048	1.000	0.000	0.999	0.001
neqr	2	4	ideal	2048	1.000	0.000	0.999	0.000
neqr	2	4	depol_0.001	2048	1.000	0.000	0.801	0.010
neqr	2	4	depol_0.005	2048	1.000	0.000	0.331	0.011
neqr	2	4	depol_0.01	2048	1.000	0.000	0.118	0.006
neqr	2	4	depol_0.05	2048	1.000	0.000	0.003	0.001
neqr	2	4	readout_0.01	2048	1.000	0.000	0.905	0.006
neqr	2	4	readout_0.05	2048	1.000	0.000	0.599	0.014
neqr	2	4	readout_0.10	2048	1.000	0.000	0.355	0.013
neqr	2	4	overrot	2048	1.000	0.000	0.972	0.003
neqr	2	4	combined	2048	1.000	0.000	0.205	0.011

geometry	neqr	qhed
circle	0.776	0.776
corner	0.750	0.750
diagonal_45	0.968	0.968
ellipse	0.895	0.895
horizontal_line	1.000	1.000
irregular_curve	0.905	0.905
multi_circles	0.905	0.905
vertical_line	1.000	1.000

geometry	method	quantum_f1	classical_f1	robinson_dominant_deg	failure_case
horizontal_line	qhed	1.000	0.800	45	no failure
horizontal_line	neqr	1.000	0.800	45	no failure
vertical_line	qhed	1.000	0.800	45	no failure
vertical_line	neqr	1.000	0.800	45	no failure
diagonal_45	qhed	0.968	0.885	0	no failure
diagonal_45	neqr	0.968	0.885	0	no failure
corner	qhed	0.750	1.000	0	A: quantum-specific
corner	neqr	0.750	1.000	0	A: quantum-specific
circle	qhed	0.776	0.938	0	A: quantum-specific
circle	neqr	0.776	0.938	0	A: quantum-specific
ellipse	qhed	0.895	1.000	45	no failure
ellipse	neqr	0.895	1.000	45	no failure
multi_circles	qhed	0.905	1.000	0	no failure
multi_circles	neqr	0.905	1.000	0	no failure
irregular_curve	qhed	0.905	1.000	45	no failure
irregular_curve	neqr	0.905	1.000	45	no failure

strategy	canny	neqr	qhed	robinson	sobel
clahe	0.564	0.806	0.787	0.709	0.677
gamma_0.5	0.000	0.729	0.195	0.769	0.781
gamma_1.5	0.486	0.724	0.589	0.767	0.771
gaussian	0.000	0.608	0.268	0.849	0.824
hist_stretch	0.610	0.726	0.711	0.772	0.781
median	0.000	0.648	0.292	0.800	0.796
none	0.199	0.726	0.428	0.770	0.781

size	method	f1_exact	f1_1px	f1_2px	precision	recall	iou	chamfer	pr_auc
8	canny	0.255	0.262	0.266	0.592	0.175	0.130	3.530	0.875
8	neqr	0.633	0.731	0.783	0.847	0.661	0.523	0.494	0.853
8	qhed	0.354	0.416	0.447	0.875	0.287	0.220	0.768	0.829
8	robinson	0.631	0.748	0.806	0.872	0.687	0.565	0.445	0.895
8	sobel	0.635	0.744	0.804	0.864	0.681	0.549	0.461	0.889
16	canny	0.353	0.383	0.397	0.814	0.264	0.184	2.929	0.715
16	neqr	0.457	0.545	0.608	0.668	0.497	0.321	1.095	0.637
16	qhed	0.126	0.155	0.175	0.619	0.092	0.061	1.992	0.574
16	robinson	0.522	0.674	0.777	0.724	0.675	0.480	0.712	0.790
16	sobel	0.536	0.690	0.792	0.723	0.700	0.497	0.692	0.792

size	method	logical_qubits	ancilla	total_qubits	raw_gates	raw_1q	raw_2q	raw_depth	transpiled_gates	transpiled_1q	transpiled_2q	transpiled_depth	statevector_MB	circuit_build_s	transpile_s	kernel_time_s	e2e_time_s	classical_e2e_s
4	qhed	4	1	5	18	12	1	8	11	10	1	5	0.00010	0.00000	0.00498	0.00268	0.18669	0.00003
4	neqr	20	2	22	116	116	0	76	78	78	0	64	67.1090	0.00023	0.00811	0.00004	0.00061	0.00003
8	qhed	6	1	7	23	15	1	9	100	58	42	84	0.00020	0.00023	0.00596	0.00298	0.22278	0.00004
8	neqr	22	2	24	622	622	0	354	358	358	0	296	268.4350	0.00107	0.00783	0.00005	0.00141	0.00003
16	qhed	8	1	9	28	18	1	10	356	187	169	333	0.00080	0.00107	0.00797	0.00379	0.27900	0.00005
16	neqr	24	2	26	2984	2984	0	1432	1438	1438	0	1184	1073.7420	0.00384	0.00784	0.00009	0.00500	0.00004

condition	Canny	NEQR-Grad	NEQR-Grad (2x2)	QPIE-QHED	Robinson	Sobel
clean	0.833	0.776	nan	0.776	0.875	1.000
directional_illumination	0.833	0.776	nan	0.776	0.941	0.952
heavy_blur_2.0	0.833	0.899	nan	0.848	0.897	0.917

condition	Canny	NEQR-Grad	NEQR-Grad (2x2)	QPIE-QHED	Robinson	Sobel
low_contrast_0.25	0.833	0.776	nan	0.776	0.875	1.000
low_illumination	0.833	0.776	nan	0.776	0.875	1.000
low_resolution_4x4	0.000	0.933	nan	0.933	1.000	1.000
quantum_combined	nan	nan	1.000	0.912	nan	nan
quantum_depol_0.05	nan	nan	1.000	0.897	nan	nan
quantum_readout_0.10	nan	nan	1.000	0.911	nan	nan
shadow_region	0.833	0.857	nan	0.857	0.919	0.957
strong_noise_0.10	0.824	0.769	nan	0.778	0.921	0.962

condition	method	severity
clean	Sobel	none
clean	Canny	none
clean	Robinson	none
clean	QPIE-QHED	none
clean	NEQR-Grad	none
low_contrast_0.25	Sobel	none
low_contrast_0.25	Canny	none
low_contrast_0.25	Robinson	none
low_contrast_0.25	QPIE-QHED	none
low_contrast_0.25	NEQR-Grad	none
heavy_blur_2.0	Sobel	none
heavy_blur_2.0	Canny	none
heavy_blur_2.0	Robinson	none
heavy_blur_2.0	QPIE-QHED	none
heavy_blur_2.0	NEQR-Grad	none
strong_noise_0.10	Sobel	none
strong_noise_0.10	Canny	none
strong_noise_0.10	Robinson	none
strong_noise_0.10	QPIE-QHED	none
strong_noise_0.10	NEQR-Grad	none
directional_illumination	Sobel	none
directional_illumination	Canny	none
directional_illumination	Robinson	none

condition	method	severity
directional_illumination	QPIE-QHED	none
directional_illumination	NEQR-Grad	none
shadow_region	Sobel	none
shadow_region	Canny	none
shadow_region	Robinson	none
shadow_region	QPIE-QHED	none
shadow_region	NEQR-Grad	none
low_illumination	Sobel	none
low_illumination	Canny	none
low_illumination	Robinson	none
low_illumination	QPIE-QHED	none
low_illumination	NEQR-Grad	none
low_resolution_4x4	Sobel	none
low_resolution_4x4	Canny	severe
low_resolution_4x4	Robinson	none
low_resolution_4x4	QPIE-QHED	none
low_resolution_4x4	NEQR-Grad	none
quantum_depol_0.05	QPIE-QHED	none
quantum_depol_0.05	NEQR-Grad (2x2)	none
quantum_readout_0.10	QPIE-QHED	none
quantum_readout_0.10	NEQR-Grad (2x2)	none
quantum_combined	QPIE-QHED	none
quantum_combined	NEQR-Grad (2x2)	none

ID	Hypothesis	Verdict
H1	Degradation reduces F1 at different rates for the two representations	CONFIRMED (Exp 4)
H2	F1 improves with shots, saturating with diminishing returns	CONFIRMED (Exp 3)
H3	QHED is weaker on curved/diagonal edges than axis-aligned ones	see Exp 6 table
H4	Low-contrast rims give significantly lower quantum F1	CONFIRMED (Exp 1 lunar, Exp 4 contrast)
H5	Equal depolarizing rates hurt deeper NEQR more	see Exp 5 table
H6	Resource cost grows far faster than the logarithmic qubit count	CONFIRMED (Exp 2, Exp 9)
H7	Preprocessing helps low-contrast lunar patches	see Exp 7 table
H8	Quantum E2E time exceeds classical at all scales	CONFIRMED (Exp 9 timing)